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Oefeningenreeks 6A nr. 3

$x_k = ak^2 + bk + c$

$x_{k+1} = a(k+1)^2 + b(k+1) + c = a(k^2 + 2k + 1) + b(k+1) + c$

$x_{k+2} = a(k+2)^2 + b(k+2) + c = a(k^2 + 4k + 4) + b(k+2) + c$

$x_{k+3} = a(k+3)^2 + b(k+3) + c = a(k^2 + 6k + 9) + b(k+3) + c$

$x_k - 3x_{k+1} + 3x_{k+2} - x_{k+3} = [ak^2 + bk + c] - 3[a(k^2 + 2k + 1) + b(k+1) + c] + 3[a(k^2 + 4k + 4) + b(k+2) + c] - [a(k^2 + 6k + 9) + b(k+3) + c]$

$ak^2 - 3a(k^2 + 2k + 1) + 3a(k^2 + 4k + 4) - a(k^2 + 6k + 9) = 0$

$bk - 3b(k+1) + 3b(k+2) - b(k+3) = 0$

$c - 3c + 3c - c = 0$

$\Rightarrow x_k - 3x_{k+1} + 3x_{k+2} - x_{k+3} = 0$ met $a, b, c \in \mathbb{R}$

References